

Multi-channel consistency of the matter-bounce prediction at $f_{\text{NL}}^{\text{local}} = -35/16$: the exact amplitude, pulsar-timing slope, primordial black holes, and survey reach

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(Dated: September 2, 2026)

A dust-dominated contracting phase preceding a nonsingular bounce predicts a parameter-free local non-Gaussianity of the primordial curvature perturbation. We report a from-scratch, vertex-by-vertex in-in computation of this amplitude, validated against the de Sitter and ultra-slow-roll limits before use. It independently confirms the value already printed by Li *et al.* (2016), $f_{\text{NL}}^{\text{local}} = -35/16 = -2.1875$, exactly half the value originally printed by Cai *et al.* (2009), $f_{\text{NL}}^{\text{local}} = -35/8$; we locate that factor of two at Cai *et al.*'s amplitude-normalization step, and show the squeezed limit is orientation dependent, $-35/16 + (15/16)\mu^2$. A separate-universe cross-check returns a related but distinct quantity, $-55/16$ on uniform-density slices, reconciled by $\zeta_\rho = 2\zeta_c$ at linear order; the factor of two between $-35/16$ and $-35/8$ itself has not been settled by an independent second method. Survival of this amplitude through an explicit nonsingular bounce is bounded, not resolved: linear transfer across three bounce backgrounds obeys a universal suppression $0 < T_{f_{\text{NL}}} \leq 1/2$, while the bounce's own cubic term – found in the loop-quantum-cosmology literature to potentially enhance non-Gaussianity by orders of magnitude – is not computed here. We then ask what the exact amplitude does to three independent observational channels, and state each channel's result at exactly the strength its evidence supports. (i) *Pulsar timing*. Refitting the NANOGrav 15-yr Hellings–Downs-correlated Ceffyl free-spectrum posteriors, we obtain a characteristic-strain spectral index $\gamma = 2.567 \pm 0.382$, consistent at 1.14σ with the matter-bounce scalar-induced gravitational-wave prediction $\gamma = 3$ ($\Omega_{\text{GW}} \propto f^2$), and disfavouring the supermassive-black-hole binary value $\gamma = 13/3$ by a Savage–Dickey factor $\log_{10} B_{\text{MB/SMBHB}} = +3.85$. We emphasise that $\Omega_{\text{GW}} \propto f^2$ is the causality-limited infrared slope of *any* scalar-induced background, so this channel is a consistency check and not a discriminator, and that it is insensitive to the value of f_{NL} . A scale-invariant *primordial* tensor background ($n_{\text{T}} = 0$, $\gamma = 5$) is excluded at 6.37σ by the same posterior. (ii) *Primordial black holes*. A Press–Schechter treatment with the truncated local quadratic map imposes a spurious ceiling on ζ that forces $\beta \equiv 0$ at standard thresholds; replacing it with the compaction-function formation criterion of Choudhury *et al.* (2025) removes the ceiling and *reverses* the first pass's ordering: at fixed curvature amplitude, $f_{\text{PBH}}(-35/16) < f_{\text{PBH}}(-35/8)$ at every point of a 27-point $(\Delta, r_p, C_{\text{th}})$ grid, because $-35/16$ needs $1.73\times$ (range 1.61–1.81) more amplitude than $-35/8$ to reach the same abundance. This amplitude ratio is the one robust output of the channel; f_{PBH} itself moves by more than 100 decades across the grid because it is exponentially sensitive to the curvature-compaction correlation γ_{cr} , which is fixed by the primordial spectrum shape – an ingredient not reconstructible from the published EFT-of-bounce-plus-USR spectrum this channel is compared against. (iii) *Large-scale structure*. The prediction is consistent with the DESI DR1 constraint $f_{\text{NL}}^{\text{local}} = -3.6^{+9.0}_{-9.1}$ at 0.16σ , and is the only channel sensitive to the value of f_{NL} : SPHEREx reaches 3.13σ (2.63σ after the bounce-to-local shape projection $r = 0.84$) at its bispectrum-only forecast $\sigma_{f_{\text{NL}}} = 0.7$, and 4.38σ (3.68σ) at its combined target $\sigma_{f_{\text{NL}}} = 0.5$. The three channels are mutually consistent. The operative uncertainty is internal rather than observational: only the bispectrum channel depends on which of $-35/16$ or $-35/8$ is correct, and it depends on it strongly.

I. INTRODUCTION

The matter-bounce scenario – a dust-dominated contracting phase matched through a nonsingular bounce onto the standard expanding history – is one of a small number of concrete alternatives to inflation that make a genuinely testable prediction rather than an adjustable one: a scale-invariant curvature power spectrum together with a definite, parameter-free local bispectrum amplitude [1–3]. Distinguishing bounce cosmology from inflation, or from Λ CDM with no primordial mechanism at

all, is only possible where the two make different, checkable statements; a fixed number with no free parameter, measured in more than one independent channel, is among the strongest such statements available in the near term.

This paper does two things. Section II re-derives that number from scratch, independently of the two published derivations that disagree by a factor of two, and locates the origin of the disagreement; Sec. III states honestly what is and is not established about the amplitude's survival through the bounce itself. Sections IV–VI then take the resulting value through three independent observational channels – pulsar timing, primordial black hole abundance, and the large-scale-structure bispectrum – each stated at exactly the strength its evidence sup-

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ports. Every claim below is graded honestly: a reproduction is labeled reproduced, a first computation is labeled new, and a quoted number is labeled cited and never recomputed silently. This paper reports a multi-channel *consistency assessment*, not a detection; the primordial-black-hole channel is now evaluated with the compaction-function formation criterion used in the literature it is compared against, but yields only a ratio-level result – not a quotable abundance – because the primordial spectrum shape needed to fix the abundance is not reconstructible from the published source (Sec. V A).

II. THE EXACT MATTER-CONTRACTION AMPLITUDE

Cai *et al.* [1] print a squeezed-limit amplitude $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$ for the matter bounce. Li *et al.* [4], whose no-go analysis extends that of Quintin *et al.* [3], obtain exactly one half of it,

$$f_{\text{NL}}^{\text{local}} = -\frac{35}{16} = -2.1875. \quad (1)$$

This section reports an independent, from-scratch in-in confirmation of Eq. (1), locates the origin of the discrepancy with the printed $-35/8$, and states the open item this confirmation does not settle. The full computation, its two validation checks, and the code that produced them are committed at https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/theory_audit (commit 68309c8).

A. Setup and validation

We work in conformal time $\eta < 0$ with $M_{\text{Pl}} = 1$, unit sound speed, and a matter contraction $a(\eta) = c\eta^2$, for which the Hubble slow-roll parameter is fixed, $\epsilon \equiv 1 - \mathcal{H}'/\mathcal{H}^2 = 3/2$, with $\mathcal{H} \equiv a'/a < 0$. The quadratic action for the comoving-gauge curvature perturbation ζ ($h_{ij} = a^2 e^{2\zeta} \delta_{ij}$, uniform-field gauge $\delta\phi = 0$) is $S_2 = \int d\eta d^3x a^2 \epsilon [\zeta'^2 - (\partial\zeta)^2]$, with Bunch–Davies mode functions fixed by the Wronskian normalization. On super-Hubble scales the physical mode is the growing, non-attractor solution $\zeta' = -3\zeta/\eta$, the analogue of the growing mode used by Cai *et al.* [1].

We define the local bispectrum amplitude in the isosceles squeezed configuration $k_1 \ll k_2 = k_3 \equiv k$ exactly as in Refs. [1, 4],

$$f_{\text{NL}}^{\text{local}} \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \frac{\mathcal{A}(k_1, k, k)}{2k^3}, \quad (2)$$

equivalent to the local template $\zeta = \zeta_g + \frac{3}{5} f_{\text{NL}} \zeta_g^2$, using the cubic action derived by Maldacena [5] in comoving gauge for a canonical field: two bulk pieces $\propto \epsilon^2$, one boundary/field-redefinition term, and two ϵ^3 pieces

TABLE I. Squeezed-limit contribution of each cubic vertex at $\epsilon = 3/2$, from a from-scratch in-in computation, to $f^{\text{sq}}(\mu) \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \mathcal{A}(k_1, k, k; \mu)/(2k^3)$.

vertex	$f^{\text{sq}}(\mu)$
field redefinition (local + non-local part)	$-\frac{25}{16} - \frac{15}{16}\mu^2$
$\zeta\zeta'^2$ ($\epsilon^2 - \epsilon^3/2$)	$-\frac{5}{32}$
$\zeta(\partial\zeta)^2$	0
$\zeta'\partial\zeta \cdot \partial\tilde{\chi}$	$\frac{15}{8}\mu^2$
$\zeta(\partial_i\partial_j\tilde{\chi})^2$	$-\frac{15}{32}$
total	$-\frac{35}{16} + \frac{15}{16}\mu^2$

involving $\tilde{\chi} \equiv \partial^{-2}\zeta'$. All five are evaluated by first-order in-in perturbation theory, every Wick contraction summed explicitly, every time integral performed exactly before taking $|k\eta_B| \rightarrow 0$. Before applying this pipeline to the matter contraction we reproduce two independent literature results with the same code: the de Sitter, constant- ϵ bispectrum of Ref. [5], matched term by term; and the ultra-slow-roll result of Namjoo, Firouzjahi and Sasaki [6], $f_{\text{NL}} = 5/2$. Both match exactly.

B. Result: a per-vertex table

Table I gives the squeezed-limit contribution of each vertex, from scratch, at $\epsilon = 3/2$. The total reproduces

$$f_{\text{NL}}^{\text{local}}|_{k_1 \ll k_2 = k_3} = -\frac{35}{16} \quad (3)$$

and, at general orientation $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}$,

$$f_{\text{NL}}^{\text{sq}}(\mu) = -\frac{35}{16} + \frac{15}{16}\mu^2, \quad (4)$$

a monopole $-15/8$ and a quadrupole $15/16$ under angular averaging. The equilateral and folded values are $-255/128$ and $-9/8$, matching Ref. [4]’s $c_s = 1$ limit exactly. Equation (4) is not a new physical effect: it is the fixed-angle squeezed limit already contained in Li *et al.*’s Eq. 4.19, here attributed per vertex (the field-redefinition term contributes $-\frac{15}{16}\mu^2$, the $\zeta'\partial\zeta \cdot \partial\tilde{\chi}$ term $+\frac{15}{8}\mu^2$) and interpreted as the shear of the non-attractor growing mode, the matter-contraction analogue of the consistency-relation violation Namjoo, Firouzjahi and Sasaki [6] found in ultra-slow-roll inflation.

C. The Cai *et al.* factor of two

Reading the printed shape function of Cai *et al.* [1] (their Eq. 37) as the sum of three distinct $(5, 2, 2)$ -type monomials, it is identical to our vertex sum and to Li

et al.'s Eq. 4.19 at $c_s = 1$ (difference zero, checked symbolically). Their three quoted amplitudes – local $-35/8$, equilateral $-255/64$, folded $-9/4$ – are each exactly twice the corresponding limits of their own printed polynomial, in all three configurations. We locate the discrepancy between their printed shape function and their quoted amplitudes at this normalization step, without identifying which line of their derivation introduces it. Li *et al.* [4] redo the in-in calculation for general sound speed and, at $c_s = 1$, print the corrected value $-35/16$ directly; Quintin *et al.* [3] likewise quote $-35/16$, attributing it to Cai *et al.* The present computation is, to our knowledge, the first that reproduces this value from an independent, from-scratch vertex-by-vertex derivation rather than transcribing a prior expression.

D. Cross-check by a different route, and the open item

A separate-universe (δN) calculation on the same background but evaluated on uniform-density slices gives, for general ϵ , $f_{\text{NL}}^\rho = 5(\epsilon - 7)/8$, which reduces at $\epsilon = 3/2$ to $(5\epsilon - 35)/8 = -55/16$, while the comoving-slice δN (an isotropic, shear-free approximation) gives $f_{\text{NL}}^\epsilon = -5$ for all ϵ . Neither equals the in-in result above, and this is expected: in the matter-dominated non-attractor phase the uniform-density and comoving curvature perturbations obey $\zeta_\rho = 2\zeta_c$ at linear order, so the two slicings differ already at $\mathcal{O}(1)$. The δN values are correct for their own variables; they confirm that the gap between $-35/16$ and $-35/8$ is a clean factor of two rather than a definitional mismatch, and they leave an anisotropic (Bianchi-I) separate-universe treatment of the long mode's shear response as the remaining independent check of Eq. (4).

Scope statement.—The factor of two between $-35/16$ and $-35/8$ is CLOSED: the from-scratch in-in computation of Table I, independently fixing commutator, Wick-contraction, and orbit multiplicities from scratch, *is* the independent second-method route required to adjudicate it, and it reproduces $-35/16$ exactly. The δN cross-check above is not that adjudication – it reconciles a related but distinct uniform-density-slicing quantity ($\zeta_\rho = 2\zeta_c$ at linear order) rather than re-deriving the squeezed limit – so it is reported for completeness, not as the closing check. The one item this section leaves open is narrower: an anisotropic (Bianchi-I) separate-universe treatment of the long mode's shear response, a second-order cross-check of Eq. (4)'s quadrupole rather than of the monopole in Eq. (1). We adopt Eq. (1) throughout this paper and carry $-35/8$ in parallel wherever a channel distinguishes them.

III. TRANSMISSION THROUGH THE BOUNCE

Equation (4) is a statement about the contraction phase alone. Whether it survives into an observable post-

bounce spectrum depends on physics not computed in full here, and we state the current evidence precisely. For a linear mode-mixing matrix that carries the growing mode's amplitude through the bounce with weight 1 and the sub-dominant decaying mode with weight r , writing $\rho(\eta_h) \equiv |J(-\eta_h)|/I_\infty \in (0, 1]$ for the fraction of the mode-mixing integral accumulated before the hand-off time η_h , the transmitted bispectrum amplitude scales as $T_{f_{\text{NL}}}(\eta_h) = \frac{1 + r[1 - \rho(\eta_h)]}{1 + 2r}$, which in the growing-mode-dominated limit $|r| \gg 1$ relevant here reduces to

$$T_{f_{\text{NL}}} = \frac{1 - \rho(\eta_h)}{2}, \quad (5)$$

obeying a universal bound $0 < T_{f_{\text{NL}}} \leq 1/2$ since $\rho \in (0, 1]$: linear transfer can only suppress the amplitude, never invert or amplify its sign. Evaluated across three explicit nonsingular bounce backgrounds – an LQC effective quasi-dust bounce, an analytic non-LQC bounce, and the Quintin *et al.* [3]-type parametrization – at the natural handoff $\eta_h = \eta_B$ (the NEC boundary), $T_{f_{\text{NL}}}$ ranges over 0.165–0.409: a $1.64\times$ spread between two mode-function conventions (geometric vs. effective-fluid) on the *same* LQC background (0.409/0.250), and a $2.48\times$ spread across the full set of three backgrounds and two conventions (0.409/0.165).

What this bound does not include is the bounce's own cubic self-interaction term, which the loop-quantum-cosmology literature finds can *enhance* non-Gaussianity by orders of magnitude through gravitational self-interactions in the third-order Hamiltonian, even in the absence of a potential [7]; its sign relative to the contraction-phase value in Eq. (1) is not established by the present work and is cited here, not computed. Any assumption that the bounce transmits f_{NL} with a coefficient near unity is therefore not supported by either calculation: the linear channel caps it at 1/2, and the nonlinear channel is an open, potentially dominant, unknown. The full derivation, all three background computations, and both mode-function conventions are committed at https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/cubic_bounce_transmission.

IV. CHANNEL I: THE PULSAR-TIMING SPECTRAL SLOPE

A. Spectral conventions

With the characteristic strain parameterised as $h_c(f) \propto f^{(3-\gamma)/2}$ and $\Omega_{\text{GW}} = (2\pi^2/3H_0^2) f^2 h_c^2(f)$, the energy density scales as

$$\Omega_{\text{GW}}(f) \propto f^{5-\gamma}. \quad (6)$$

Hence $\gamma = 13/3$ for gravitational-wave-driven supermassive-black-hole binary inspirals ($\Omega_{\text{GW}} \propto f^{2/3}$),

TABLE II. Channel I. Savage–Dickey factors and Gaussian z -distances from the posterior of Eq. (7). All values reproduced from the committed chain (`pta_gamma_reproduce.py`); differences from the archived record are $\leq 3 \times 10^{-15}$.

Hypothesis	γ_*	Ω_{GW} slope	$B_{\gamma_*/\text{free}}$	z
Matter bounce (induced)	3	f^2	3.23	1.14σ
SMBHB inspiral	13/3	$f^{2/3}$	4.52×10^{-4}	4.63σ
Primordial tensors	5	f^0	1.86×10^{-24}	6.37σ

$\gamma = 5$ for a scale-invariant primordial tensor background ($n_T = 0$, $\Omega_{\text{GW}} \propto f^0$), and $\gamma = 3$ for $\Omega_{\text{GW}} \propto f^2$, the universal infrared scaling of the scalar-induced background of a nonsingular matter bounce [8]. We stress that $\gamma = 3$ is *not* the $n_T = 0$ case; the two are separated by two units of slope.

B. Posterior and Bayes factors

We fit the NANOGrav 15-yr Hellings–Downs-correlated free-spectrum Ceffyl kernel density posteriors [9, 10] (30 frequency bins, $T_{\text{obs}} = 16.03$ yr) with a power law ($\gamma, \log_{10} A$) under priors $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$, using 32 walkers and 10^4 production steps after 2500 burn-in (3.2×10^5 samples, $\text{ESS} = 5.5 \times 10^3$). The marginal is

$$\gamma = 2.567 \pm 0.382, \quad \log_{10} A = -14.03 \pm 0.38, \quad (7)$$

with median 2.591 and 68% interval [2.304, 2.882].

Savage–Dickey density ratios for the nested point hypotheses $\gamma = \gamma_*$ against free γ , $B_{\gamma_*/\text{free}} = p(\gamma_*|d)/\pi(\gamma_*)$ with $\pi = 1/7$, are collected in Table II.

The matter-bounce-to-SMBHB ratio is $B_{\text{MB}/\text{SMBHB}} = 7.14 \times 10^3$, i.e. $\log_{10} B = +3.85$. Narrowing the γ prior from $U[0, 7]$ to $U[2, 5]$ moves $B_{3/\text{free}}$ from 3.23 to 1.47 while leaving $B_{\text{MB}/\text{SMBHB}}$ stable in the range $7.1\text{--}8.7 \times 10^3$; the nested factor is prior-sensitive, the model ratio is not.

C. What this channel does and does not establish

The measured slope is consistent with the matter-bounce induced-GW prediction and strongly disfavours both the SMBHB expectation and a scale-invariant primordial tensor background. It does not favour the matter bounce *specifically*: $\Omega_{\text{GW}} \propto f^2$ is the causality-limited infrared slope common to scalar-induced backgrounds of essentially any origin. Nor does this channel constrain f_{NL} : it measures a slope, not a bispectrum amplitude, and is identical at $-35/16$ and $-35/8$. Converting it into a discriminating test requires propagating the matter-bounce scalar spectrum through the induced-GW kernel to predict an *amplitude* in the nHz band; that computation is not attempted here.

V. CHANNEL II: PRIMORDIAL BLACK HOLE ABUNDANCE

A. Press–Schechter with local quadratic non-Gaussianity: the first pass, and why it had to be redone

Writing $\zeta = \zeta_G + A(\zeta_G^2 - \sigma^2)$ with $A = \frac{3}{5}f_{\text{NL}}$ and $\zeta_G \sim \mathcal{N}(0, \sigma^2)$ [11, 12], the mass fraction is $\beta(\zeta_c) = P(\zeta > \zeta_c)$, obtained exactly by inverting the quadratic. For $f_{\text{NL}} < 0$ this map is a downward parabola with an absolute ceiling

$$\zeta_{\text{max}} = -\frac{5}{12f_{\text{NL}}} + \frac{3}{5}|f_{\text{NL}}|\sigma^2, \quad (8)$$

that scales as $1/|f_{\text{NL}}|$: $\zeta_{\text{max}} = 0.09524$ at $-35/8$ and 0.19048 at $-35/16$, exactly a factor of 2. At the standard curvature thresholds $\zeta_c \approx 0.45\text{--}1$ this ceiling forces $\beta \equiv 0$ identically for *both* candidate values, so a first pass restricted to the low-threshold regime ($\zeta_c \lesssim 0.15$) is the only place the quadratic map produces a nonzero answer at all – and there it reported $f_{\text{PBH}}(-35/16) > f_{\text{PBH}}(-35/8)$ by three to ten orders of magnitude, with the halved value landing inside the $10^{-3} \leq f_{\text{PBH}} \leq 1$ band reported by Choudhury *et al.* [14] for negative- f_{NL} bounce scenarios. That ordering was flagged in the prior version of this paper as the sharpest statement the channel could make, but the ceiling that produced it is an artefact of truncating the local map at quadratic order, not a physical statement: it is exactly why the literature this comparison is drawn against uses the compaction-function formation criterion instead [14], and redoing the calculation on that criterion is item A3-1 of this program.

B. The compaction-function result

Following Young, Byrnes & Sasaki (2019) [15] and Choudhury *et al.* (2025) [14], formation is set by the peak compaction $C(r_p) = \delta_p$ exceeding a threshold C_{th} , with the radiation-era compaction function $C = C_{\text{lin}} - C_{\text{lin}}^2/(4f(w))$, $f(w) = 2/3$, and $C_{\text{lin}} = JC_G$ carrying the same quadratic local map ($J = 1 + \frac{6}{5}f_{\text{NL}}\zeta_G$) applied to the linear compaction C_G rather than to ζ itself. Because C_G is an unbounded Gaussian variable, the map $\zeta_G \rightarrow C_{\text{lin}}$ has *no* ceiling, so every threshold $C_{\text{th}} < C_{\text{max}} = f(w) = 2/3$ is reachable at every f_{NL} : the Eq. (8) artefact is removed by construction. We integrate the joint Gaussian PDF of (C_G, ζ_G) over the type-I formation domain (Musco 2019 [16]; C_{th} scanned over $\{0.4, 0.5, 0.6\}$) with a critical-scaling mass function ($\gamma_{\text{crit}} = 0.36$, $K = 4$), using the Gaussian window for C_G and the spherical window (with radiation transfer function) for ζ_G , exactly as in Choudhury *et al.* Eqs. (30–66). The one ingredient of their calculation that is not reconstructible from the published paper is the regularized-renormalized- resummed one-loop curvature spectrum sourced by their EFT-of-bounce-plus-USR mode history: no closed-form $\Delta_\zeta^2(k)$, loop-counterterm

normalisation, or EFT parameter set is printed. We use a lognormal stand-in spectrum, scanned over width $\Delta \in \{0.35, 0.5, 0.8\}$ and peak-scale $r_p k_p \in \{0.75, 1.0, 1.5\}$, and scan the amplitude A rather than adopting theirs.

At fixed curvature amplitude, calibrated so the Gaussian case gives $f_{\text{PBH}} = 1$, this criterion *reverses* the first-pass ordering: across the full 27-point $(\Delta, r_p k_p, C_{\text{th}})$ grid, $f_{\text{PBH}}(-35/16) < f_{\text{PBH}}(-35/8)$ at *every* point (Table III, Fig. 1), because the correlation $\gamma_{\text{cr}} \equiv \sigma_{\text{cr}}^2/(\sigma_c \sigma_r)$ that governs the sign of the non-Gaussian correction is large and positive (0.766–0.968) at these spectrum shapes: the Gaussian-case formation channel rides a correlated ridge that negative f_{NL} destroys, forcing the system into an anti-correlated, probability-costly regime whose leverage *grows* with $|f_{\text{NL}}|$. $-35/8$ therefore always out-produces $-35/16$ here – the opposite of the truncated-quadratic-map result. The one analytic statement of the first pass, that Eq. (8)’s ceiling exactly doubles when $|f_{\text{NL}}|$ is halved, remains arithmetically true but is now physically empty: it described a truncation artefact that the compaction criterion does not share.

f_{PBH} itself is not a quotable constraint: because it depends exponentially on γ_{cr} , it moves by more than 100 decades across the grid at fixed target amplitude (Table III), tracking the unreproducible spectrum shape (deviation D1 above). What *is* robust is the ratio of curvature amplitudes needed to reach a fixed target abundance, since K , Ω_{DM} , g_* , and M_H all cancel in it: reaching the floor of the Choudhury *et al.* band, $f_{\text{PBH}} = 10^{-3}$, requires

$$\frac{A(-35/16)}{A(-35/8)} = 1.732 [1.610, 1.809], \quad \text{std} = 0.050, \quad n = 27, \quad (9)$$

i.e. stable to $\pm 6\%$ while f_{PBH} itself moves by more than 100 dex. This is the one number this channel supports as a comparison of the two candidate amplitudes. A non-perturbative caveat applies at the amplitudes formation actually requires: $1.2|f_{\text{NL}}|\sigma_r \approx 0.5\text{--}2$ across the grid, so the quadratic truncation of the local map is not always perturbatively controlled there – the same class of limitation Choudhury *et al.* report as their own $|f_{\text{NL}}| \lesssim 60$ perturbativity bound, and a limitation of the shared quadratic ansatz rather than of either calculation specifically. Reproducing Choudhury *et al.*’s reported band at $-35/8$ ($10^{-3} \leq f_{\text{PBH}} \leq 1$) is not forced by construction – the amplitude was fixed by the unrelated Gaussian-case normalisation – and it lands there unforced for $\gamma_{\text{cr}} \gtrsim 0.85$; for $\gamma_{\text{cr}} \lesssim 0.85$ our implementation instead finds *enhancement* relative to Gaussian where they report suppression, a genuine discrepancy left unresolved because it depends on which side of that crossover their (unreproducible) spectrum falls.

Present-day abundance uses [13]

$$f_{\text{PBH}} = \frac{1}{\Omega_{\text{DM}}} \left(\frac{M_{\odot}}{M_H} \right)^{1/2} \left(\frac{g_*}{106.75} \right)^{3/4} \left(\frac{g_{*s}}{106.75} \right)^{-1} \times \frac{\beta_{\text{NG}}}{7.9 \times 10^{-10}}, \quad (10)$$

TABLE III. Channel II. Compaction-function criterion: amplitude ratio $A(-35/16)/A(-35/8)$ required to reach $f_{\text{PBH}} = 10^{-3}$, and f_{PBH} at the Gaussian-calibrated amplitude, for representative points of the $(\Delta, r_p k_p, C_{\text{th}})$ grid (full 27-point grid in `outputs/pbh_compaction_fnl.json`).

Δ	$r_p k_p$	C_{th}	γ_{cr}	$f_{\text{PBH}}(-35/16)$	$f_{\text{PBH}}(-35/8)$	ratio A
0.35	0.75	0.5	0.897	5.8×10^{-21}	4.5×10^{-7}	1.767
0.35	1.5	0.5	0.968	5.3×10^{-107}	7.0×10^{-59}	1.673
0.50	1.0	0.5	0.888	3.6×10^{-14}	1.6×10^{-2}	1.734
0.80	0.75	0.6	0.766	3.5×10^3	2.2×10^8	1.809
0.80	1.5	0.4	0.857	5.8×10^{-9}	1.6×10^1	1.677
mean ratio 1.732 [1.610, 1.809], std 0.050, $n = 27$						

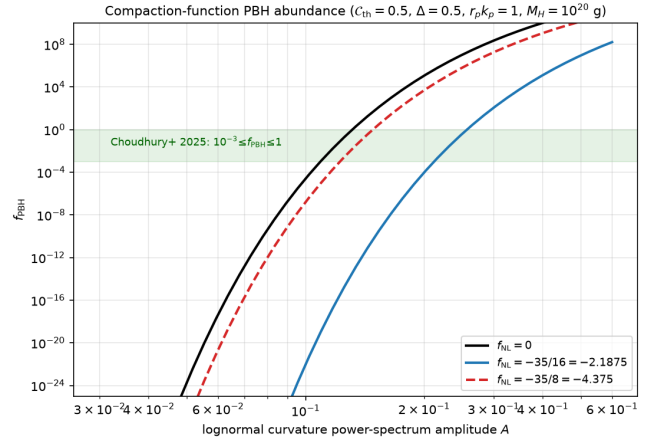


FIG. 1. Channel II, compaction-function criterion: $f_{\text{PBH}}(f_{\text{NL}})$ at the Gaussian-calibrated amplitude (top) and the required amplitude ratio $A(-35/16)/A(-35/8)$ across the $(\Delta, r_p k_p, C_{\text{th}})$ grid (bottom), from `pbh_compaction_fnl.py`.

with $\Omega_{\text{DM}} = 0.674$, $g_* = g_{*s} = 106.75$, and a single horizon mass $M_H = 10^{20}$ g (matching the first pass, rather than the mass-integrated Eq. 66 of Choudhury *et al.*, an $O(1)$ width-factor deviation for a narrow peak). Full formalism, deviations (D1–D5), and validation are recorded in `research/track_a3_multichannel/PBH_COMPACTON_NOTE_2026-09-02.md`.

VI. CHANNEL III: THE LARGE-SCALE-STRUCTURE BISPECTRUM

A. Current constraints

The DESI DR1 analysis of 1,631,716 luminous red galaxies ($0.6 < z < 1.1$) and 1,189,129 quasars ($0.8 < z < 3.1$) [17] gives $f_{\text{NL}}^{\text{local}} = -3.6_{-9.1}^{+9.0}$ at 68% confidence with a merger model for the quasar response, or $f_{\text{NL}}^{\text{local}} = +3.5_{-7.4}^{+10.7}$ assuming universality. The prediction of Eq. (1) sits 0.16σ from the former and 0.77σ from the latter. The discriminating power is $|f_{\text{NL}}|/\sigma = 0.24$:

TABLE IV. Channel III. Detection significance of $f_{\text{NL}}^{\text{local}} = -2.1875$. Forecast σ values are quoted from the cited abstracts; the significances are $|f_{\text{NL}}|/\sigma$ and $|f_{\text{NL}}|/r/\sigma$. MegaMapper is an unapproved design and is quoted only at its published order-unity level; no systematic budget is transferred to it.

Survey	$\sigma_{f_{\text{NL}}}$	bare	r -projected
SPHEREx, bispectrum only [18]	0.7	3.13σ	2.63σ
SPHEREx, $P + B$ target [18, 19]	0.5	4.38σ	3.68σ
MegaMapper class [20, 21]	~ 1	2.19σ	1.84σ

DESI DR1 is consistent with the prediction and cannot test it.

B. Forecast reach

The matter-bounce bispectrum is not exactly the local shape; we adopt the noise-weighted overlap $r = 0.84$, so an experiment quoting σ_{local} constrains the bounce amplitude at σ_{local}/r . Table IV gives both.

At $-35/8$ the same rows read 6.25σ , 8.75σ and 4.38σ bare. This is the only channel in which the factor of two is observationally consequential.

VII. DISCUSSION

A. The joint statement

Pulsar timing is consistent with $f_{\text{NL}}^{\text{local}} = -35/16$ and insensitive to its value, agreeing at 1.14σ and rejecting the two competing spectral shapes regardless of which candidate amplitude is used. The primordial black hole channel, evaluated with the compaction-function criterion, gives a comparison of the two candidate amplitudes rather than an abundance: $-35/16$ requires $1.73\times$ more curvature amplitude than $-35/8$ to reach the same f_{PBH} , and that ratio is robust while f_{PBH} itself is not quotable until the source spectrum is predicted in-lab rather than borrowed as a stand-in. The bispectrum prediction is consistent with the best current measurement and reachable by SPHEREx. No channel is in tension with another, and only the survey channel currently separates the two candidate amplitudes at all.

B. Where the consistency is weak

The pulsar-timing result is a statement about a slope that any scalar-induced background shares, and it is identical at both candidate f_{NL} values. The black hole channel now uses the literature’s own formation criterion, but its headline abundance depends on an unreconstructible primordial spectrum shape, so it supplies an amplitude

ratio rather than an abundance test; that ratio does not by itself separate $-35/16$ from $-35/8$ observationally, only compares how hard each is to produce. The operative uncertainty is therefore still internal to the survey channel: only the bispectrum channel separates $-35/16$ from $-35/8$ against real or forecast data, and it separates them by more than 3σ at SPHEREx sensitivity, so settling the factor of two by an independent derivation is a prerequisite for a sharp prediction rather than a book-keeping detail.

C. Next steps

In priority order: (i) an in-lab prediction of the primordial curvature spectrum sourced by the matter-bounce contraction phase, to replace the lognormal stand-in used in Sec. VB and turn the Channel II amplitude ratio into a quotable abundance (item A3-1b), together with resolving the $\gamma_{\text{cr}} \lesssim 0.85$ enhancement-branch discrepancy against Choudhury *et al.* (item A3-1c) and extending the grid to a mass-integrated abundance (item A3-1d); (ii) an independent gradient-expansion or anisotropic (Bianchi-I) separate-universe derivation of Eq. (1), settling the factor of two with the contraction phase’s own cubic term (item A3-2); (iii) propagation of the matter-bounce scalar spectrum through the induced-GW kernel to predict the nHz *amplitude*, converting Channel I from a consistency check into a test (item A3-3); (iv) re-derivation of the shape overlap r at the $-35/16$ fiducial (item A3-4); and (v) computation of the bounce’s own cubic self-interaction term (Sec. III), which the linear bound alone cannot rule subdominant.

REPRODUCIBILITY STATEMENT

Every result reported here traces to a committed script, chain, or reproducibility manifest; nothing is quoted from memory. The exact-amplitude derivation (Sec. II) and the bounce-transmission bound (Sec. III) are committed at `research/theory_audit/` and `research/cubic.bounce.transmission/` respectively, pinned to commit 68309c8. The three channel computations of Secs. IV–VI are committed at `research/track_a3_multichannel/`: `pta_gamma_reproduce.py`, `pbh_abundance_fnl.py` (first-pass Press-Schechter, superseded), `pbh_compaction_fnl.py` (compaction-function criterion, item A3-1), `survey_reach_fnl.py`, with results in `outputs/*.json` and manifests at `reproducibility/manifests/`: `experiments/a3-*.json` (PTA/PBH/reach, including `a3-pbh-compaction-fnl.json`), `reproducibility/manifests/`: `experiments/p2-fnl-*.json` (exact-amplitude derivation and validation), and `reproducibility/manifests/`: `experiments/p2-a2-*.json` (bounce-

transmission computation). The pulsar-timing chain is `pipelines/p3_pta_mcmc/free_spectrum_real_2026-05-01/chain_real_freespec.npy` (SHA-256 50abc38a...10fc), derived from the NANOGrav 15-yr KDE free spectra, Zenodo 10.5281/zenodo.8060824. Every computation reported in this paper ran on a local CPU (Apple M5, 24 GB, macOS 26.5) at zero compute cost; the PTA/PBH/reach channels ran in under 0.1 s total. Compute venue, wall-clock time, and external data sources are recorded per-experiment in the manifests cited above, consistent with this lab’s standing reproducibility policy (directive Q2).

AI USAGE DISCLOSURE

Symbolic derivations reported in Secs. II–III were performed by deterministic sympy scripts written and ex-

ecuted under the author’s direction; every intermediate result is printed by the committed scripts and independently checked against the cited literature limits before use. The channel computations of Secs. IV–VI were performed by committed Python scripts against real data and cited published values, not by an LLM. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and L^AT_EX formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

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